PHP Programming

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Sekilas tentang PHP

- Bahasa pemrograman script di sisi server
- Berupa script yang disisipkan di dalam dokumen HTML, *embedded script* yang diinterpretasi (bukan di-compile)
- PHP singkatan dari:
  - **P**ersonal **H**ome **P**age
  - **P**rofessional **H**ome **P**age
  - **P**HP: **H**ypertext **P**reprocessor
- Official Website : http://www.php.net
- PHP versi terakhir : 4.3.10 & 5.0.3
- Free & opensource
- Multi platform: Windows, Linux, Mac
- Menyediakan Library/API yang menyeluruh:
  - Database : MySQL, Oracle, postgreSQL, IBM DB2, ODBC, dll
  - Protocol : HTTP, FTP, POP3, IMAP, LDAP, SNMP, dll
  - Output : HTML, XML, JPEG, GIF, PNG, PDF, SWF
  - dll

Contoh Penyisipan Script PHP

PHP

```
<html>
<?
  /* Cara I: script PHP dideklarasikan di sini*/
  echo "halo 1<br>";
?>

<?php
  /* Cara II: script PHP dideklarasikan di sini*/
  echo "halo 2<br>";
?>

<script language="php">
  /* Cara III: script PHP dideklarasikan di sini*/
  echo "halo 3";
</script>
</html>
```

HTML

```
<html>
halo 1<br>
halo 2<br>
halo 3</html>
```

Browser

halo 1
halo 2
halo 3

Identifier, Tipe, Variabel, Konstanta

- case sensitive
- variable dengan prefiks: `$`
- variable tidak perlu dideklarasi
- *weakly typed*, dengan tipe data:
  - Jenis scalar : integer, float/double, string, boolean
  - Jenis compound : array (indexed, associative), object
  - Jenis khusus : resource, NULL
- pendefinisian tipe:
  - fungsi settype: `settype($var,"integer");`
  - fungsi konversi: `intval($var), floatval($var), strval($var)`
  - type casting: `(integer)$var, (float)$var, (string)$var`
  - type juggling:

```
$foo = "0";                          // $foo is string (ASCII 48)
$foo += 2;                           // $foo is now an integer (2)
$foo = $foo + 1.3;                   // $foo is now a float (3.3)
$foo = 5 + "10 Little Piggies";      // $foo is integer (15)
```

- variable scope:
  - local (default)
  - global: `global $var; $GLOBALS["var"];`
  - static: `static $var;`
- konstanta
  - Pendefinisian: `define("pi", 3.14);`
  - Konstanta terdefinisi: `PHP_VERSION`, `PHP_OS`, `__FILE__`, `__LINE__`, dll

Operator

- Assignment: `=`, `+=`, `-=`, `*=`, `/=`, `%=`
- Aritmatika: `+`, `-`, `*`, `/`, `%`
- Operator pre/post increment/decrement: `++$a`, `--$a`, `$b++`, `$b--`
- Perbandingan: `==` (equal), `===` (identical), `!=`, `>`, `<`, `>=`, `<=`
- Operator logika: `&&` (and), `||` (or), `!` (not)
- Bitwise: `&` (and), `|` (or), `~` (not), `^` (xor), `<<` (shift left), `>>` (shift right)
- Operator kondisi: `?`

```
$jenis = ($bil%2==0) ? "Genap" : "Ganjil";
```

- Operator string: `.` (concat)
- Error Control:

```
$my_file = @file("data.txt") or die("Can't open file");
```

- Operator eksekusi: `$output = `ls -l`;`
- Operator new (object): `new`

Konstruksi dasar program PHP

- Sequence
- Pemilihan
  - if
  - if .. else, elseif
  - switch .. case, break
- Pengulangan
  - while
  - do .. while
  - for
  - foreach
    - foreach (array_expression as $value) <statement>
    - foreach (array_expression as $key => $value) <statement>
- Pencabangan
  - break
  - continue

Fungsi

- Passing parameter by value:

```
function increment($number) {…};
```

- Passing parameter by reference:

```
function increment(&$number) {…};
```

- Dynamic Function Call:

```
<? function printItalic($txt) {
     echo "<I>". $txt . "</I> <BR>";
   }
   function printBold($txt) {
    echo "<B>". $txt . "</B> <BR>";
   }
   $printFunction = "printItalic";
   $printFunction("Italic");  //<I>Italic</I> <BR>
   $printFunction = "printBold";
   $printFunction("Bold");    //<B>Bold</B> <BR>
?>
```

- Dynamic Function Creation:

```
<? $newfunc = create_function('$a,$b','return "$a + $b = ".($a + $b);');
   echo $newfunc(2,3)."\n";   //2 + 3 = 5
?>
```